Fluorescence Microscopy Imaging

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SUSTech-BIO 2026 iGEM Team

Purpose

Observe and document visible fluorescence and cell morphology in prepared samples, principally 293T and CCF-STTG1 cells, using a Nikon ECLIPSE Ti2 microscope and NIS-Elements. Use brightfield, EGFP, mCherry and BFP imaging as required by the reporter present.

This procedure supports qualitative checks of fluorescence presence and simple visual comparisons. It does not produce fluorescence-intensity measurements, positive-cell percentages or RNA-editing efficiency.

Materials & Equipment

Item	Provider / Identifier	Use / Specification
Prepared cell samples	Laboratory stocks; mainly 293T and CCF-STTG1	Suitable culture plates, dishes or prepared specimens; select the imaging time for the experiment.
Inverted microscope	Nikon ECLIPSE Ti2	Configured for transmitted-light and epifluorescence observation, with an attached camera. [1]
Objectives	Installed 4×, 20× and 40× objectives	4×: field finding and quick fluorescence screening; 20×: local regions; 40×: individual-cell detail.
Imaging software	Nikon NIS-Elements	Live camera view, AE, supported color controls, image capture and scale-bar annotation. Controls depend on the connected camera. [2,3]
Illumination and filters	Installed microscope configuration	Brightfield plus filter/light combinations matched to EGFP, mCherry and the actual BFP variant. [4]
Background/control samples and calibration	Matched reporter-negative cells; stage micrometer if needed	Check background; a known fluorescent sample helps investigate absent signal. Use established pixel calibration for scale bars. [5]

Before You Start

Identify the sample, well and imaging endpoint. Keep live cells covered with compatible culture or imaging medium. If samples require fixation or staining, complete that preparation beforehand; it is outside this imaging procedure.

Start the microscope, camera and NIS-Elements using the local instrument startup procedure. Check that the sample holder fits the vessel and that the required reporter channels are available. Use the actual objective's immersion and working-distance requirements.

Decide whether images are only for detecting visible fluorescence or also for a visual comparison. For comparison, use the same cell type, preparation, time point and objective, and keep acquisition and display settings consistent within each fluorescence channel.

Procedure

A. Find the field and focus

1. Place the prepared plate or dish securely on the stage and position the intended well over the objective. Use the 4× objective and transmitted light to locate cells and focus on the cell layer.

2. Inspect several areas of the well and choose a field that represents its general appearance. Briefly inspect the relevant fluorescence channel at 4× for a quick signal check. Revisit weak or sparse signals at higher magnification before concluding that no fluorescence is visible.

3. Switch to 20× for a local region or 40× for individual-cell detail, as needed. Refocus after changing objectives; use fine focus at higher magnification. Keep the field position unchanged when acquiring corresponding channels.

B. Route the image to the camera and acquire

4. Switch the microscope output to the camera port and open the live camera view in NIS-Elements. Check and refine focus on the screen. The camera light path and the fluorescence channel are separate selections.

5. Select brightfield or the installed optical configuration/filter and excitation light appropriate for EGFP, mCherry or BFP. For fluorescence, switch off transmitted illumination and open the fluorescence illumination path. Match the filters to the actual fluorescent protein rather than relying only on a displayed channel color. [4]

6. For routine visibility checks, use AE (Auto Exposure) to obtain a useful live image. Inspect the captured image for bright clipped regions or excessive background; reduce exposure or illumination if needed. Record the resulting exposure and gain. AE can change settings between fields. [2,3]

7. Where a color camera provides Auto White / AWB, apply it to a neutral brightfield scene if color correction is needed. Keep the established color settings for fluorescence; do not white-balance a colored fluorescence field. For monochrome acquisition, assign a consistent display color to each channel. [2,5]

8. For a simple brightness comparison, establish a usable, unsaturated setting on a representative bright sample and then turn off continuous AE or use AE Lock if available. Keep exposure, gain, illumination and display settings the same within that channel across the comparison. Images individually adjusted by AE remain visibility records, not evidence of relative brightness. [2,6]

9. Capture the focused image, then acquire the corresponding brightfield and required fluorescence channels from the same field. Check the saved frame before moving to another field or changing magnification.

C. Add a scale bar and save

10. Check that the selected software objective and image pixel calibration match the actual optical configuration, including any intermediate magnification and camera sampling. If calibration is uncertain, establish it with a stage micrometer before adding a physical scale bar. [5]
11. Preserve the acquired image without a burned-in annotation, then add a readable scale bar with its length and unit to an export copy. Position it away from the cells of interest. Verify its appearance in the exported file.
12. Save the original in the supported native format or lossless TIFF, retaining available metadata, and save an annotated copy for presentation. Include sample/well ID, date, imaging time point, field, channel and objective in the filename or accompanying record; retain exposure, gain and automatic-setting status.
13. Close the fluorescence illumination when imaging is finished and remove the sample. Return continuing live cultures promptly to their culture conditions, then follow the local microscope shutdown procedure. Limit unnecessary excitation during focusing and between captures. [7]

p.s.

Use 4× to navigate, 20× to inspect a region and 40× to examine a cell. A negative-looking 4× field alone is a limited screen; check other fields and the appropriate higher magnification.

Keep brightfield and fluorescence images paired by field. For visual comparisons, select fields by the same rule and retain the originals rather than keeping only the brightest cells.

AE adjusts exposure; Auto White adjusts color balance. Neither is a biological normalization. Label independently adjusted images as qualitative observations and keep any presentation adjustments consistent across images being compared. [2,6]

Recheck the scale-bar calibration after changing the optical configuration. A 40× label by itself does not specify the physical size of an image pixel. [5]

Expected Outcome

Obtain focused brightfield images with corresponding reporter-channel images and correctly calibrated scale bars. Reporter-negative cells help distinguish visible signal from background; a known fluorescent sample helps assess whether a missing signal could reflect the imaging setup.

Describe observations as visible fluorescence, no signal distinguishable from background under the recorded settings, or a qualitative difference under matched settings. Fluorescence alone does not establish RNA editing, and these images are not used to calculate positive-cell fractions or editing efficiency.

Troubleshooting

Observed problem	Check first	Action
Eyepiece image is visible; camera view is dark	Camera connection, live view, output-port selector and light path	Route light to the camera and verify the selected illumination/filter before changing the sample.
Cells are blurred or hard to find	Well position, cell-layer focus, seated objective and vessel-bottom compatibility	Relocate cells at 4×, refocus, then return to the required objective and refine focus on the camera view.
Little or no fluorescence is visible	Correct reporter/filter combination, excitation path, exposure and background control	Check a known fluorescent sample if available, inspect other fields and higher magnification, and report signal as undetectable under the recorded settings if unresolved. [4]
Image is washed out, noisy or changes brightness between fields	Saturation, AE activity, exposure, gain and display scaling	Reduce clipping or excessive gain. For comparison, reacquire with matched settings rather than interpreting independently adjusted brightness. [2,3,6]
Fluorescence fades during observation	Time spent under excitation and repeated captures	Close excitation between captures and use a fresh field with less unnecessary illumination. [7]
Scale bar is missing or inconsistent	Active objective, image calibration and annotation export	Correct calibration and export a new annotated copy from the preserved original. Check the exported bar and unit. [5]

Reference

[1] Nikon Instruments. ECLIPSE Ti2 Series: inverted microscope imaging platform. Accessed 2026-10-13. Open source

[2] NIS-Elements online help, Camera Settings: AE Lock, gain and Auto White Balance. Camera-dependent controls. Accessed 2026-10-13. Open source

[3] NIS-Elements online help, Cameras: automatic exposure, camera sensitivity and available controls. Accessed 2026-10-13. Open source

[4] Nikon MicroscopyU. Imaging Fluorescent Proteins: microscope parameters and matching filter combinations. Accessed 2026-10-13. Open source

[5] NIS-Elements online help, Calibration: channel setup, optical configurations, objectives and objective recalibration. Accessed 2026-10-13. Open source

[6] Nikon. NIS-Elements Viewer User's Guide, ver. 5.21.00, printed pp. 6-8: LUTs, auto scaling and oversaturation display. Used for display principles, not acquisition instructions. Accessed 2026-10-13. Open source

[7] Nikon MicroscopyU. Photobleaching: prolonged excitation and loss of fluorescence. Accessed 2026-10-13. Open source